

Autonomous Forest Drone

User Manual

Elina Rosato, Danielis Maizelis
Kristianstad University, Sweden

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1 Introduction

This manual covers the setup, operation, and maintenance of the Autonomous Forest Drone system developed as a bachelor's thesis in Computer Science and Engineering with a specialisation in IoT at Kristianstad University, Sweden.

The drone is designed for autonomous navigation in forest environments. **This document assumes the drone is already assembled.** It focuses on software configuration, pre-flight preparation, operation, and troubleshooting.

Intended users:

- Research operators and field technicians (primary)
- Academic supervisors and industry partners (secondary)
- Future developers extending the system (tertiary)

Users are expected to have a basic understanding of drone operation and familiarity with command-line interfaces. Some knowledge of ROS and Python is helpful but not required.

2 System Overview

2.1 Purpose

The system performs autonomous waypoint-based navigation in forest environments. Missions are scripted and executed by a Jetson Orin Nano companion computer, which communicates with the Pixhawk flight controller via MAVLink. During flight, the drone captures photos and sensor data for post-mission analysis in support of forest monitoring research.

2.2 Main Capabilities

- Autonomous waypoint navigation in forested terrain
- Photo and sensor data capture during flight
- Forest monitoring missions up to 15 minutes
- Real-time monitoring in QGroundControl
- Telemetry streaming to ground control station (CLI custom implementation)
- GPS-based positioning with barometric altitude hold
- Manual RC override available at any time
- Geofencing and automatic safety features (Return to Launch, low-battery auto-land)
- Data logging for post-flight analysis

2.3 Known Limitations

- **Flight time:** approximately 12–15 minutes (4S 5000 mAh LiPo)
- **Maximum speed:** 3 m/s (safety limit in forest environments)
- **GPS required:** does not operate indoors or under dense canopy
- **Weather:** no rain; wind must be below 3 m/s; temperature 0–35°C
- **Minimum GPS satellites:** 12 or more required before flight

2.4 Physical Specifications

Parameter	Value
Frame	Holybro S500 V2 (500 mm diagonal wheelbase)
Dimensions	550 mm × 550 mm × 200 mm (with propellers)
Total weight	approximately 2.3 kg (including battery)
Battery	4S LiPo 14.8 V 5000 mAh (XT60 connector)
Flight time	12–15 minutes
Additional payload capacity	approximately 500 g
Operating temperature	0–35°C
Maximum humidity	below 80% (no rain)
Maximum wind	3 m/s sustained
Maximum altitude	50 m AGL

3 Components Reference

This section lists all hardware components with their exact names and models. The reference numbers in the tables correspond to the labels in Figure 1. Use this section as a reference when identifying or ordering replacement parts.



Figure 1: Drone component reference diagram

3.1 Flight Platform

#	Component	Part Name / Model	Notes
1	Frame	Holybro S500 V2 Quadcopter Kit [7]	500 mm diagonal
2	Motors ($\times 4$)	Holybro 2216 880KV Brushless Motor	4 required
3	ESCs ($\times 4$)	40 A Electronic Speed Controller	Included in S500 V2 kit
4	Propellers CW ($\times 2$)	1045 Propeller (10-inch, 4.5-pitch, CW)	Unmarked or marked “R”
5	Propellers CCW ($\times 2$)	1045 Propeller (10-inch, 4.5-pitch, CCW)	Marked with “L”
6	Power distribution	Holybro PDB (Power Distribution Board)	Included in kit

3.2 Flight Controller System

#	Component	Part Name / Model	Notes
7	Flight controller	Holybro Pixhawk 6C	Main autopilot
8	GPS module	Holybro M10 GPS with compass	Dual GNSS (GPS + GLONASS)
9	Telemetry radio	RFD900x or SiK Radio	915 MHz or 433 MHz
10	Power module	Holybro PM02	Voltage and current sensing
11	RC receiver	FlySky FS-iA6B	PPM mode, 6 channels
12	RC transmitter	FlySky FS-i6	6-channel, 2.4 GHz AFHDS

3.3 Companion Computer

#	Component	Part Name / Model	Notes
13	Companion computer	NVIDIA Jetson Orin Nano 8GB	Main onboard computer
14	Camera	Arducam IMX219	8 MP, 62° FOV, CSI interface, fixed focus

3.4 Power System

#	Component	Part Name / Model	Notes
15	Main battery	4S LiPo 14.8 V 5000 mAh	XT60 connector
16	Y-splitter	XT60 Y-Splitter Cable	Battery $\rightarrow$ PDB + Jetson power
17	Battery charger	4S LiPo Balance Charger	Balance charging required

3.5 Exploded View and Connector Map

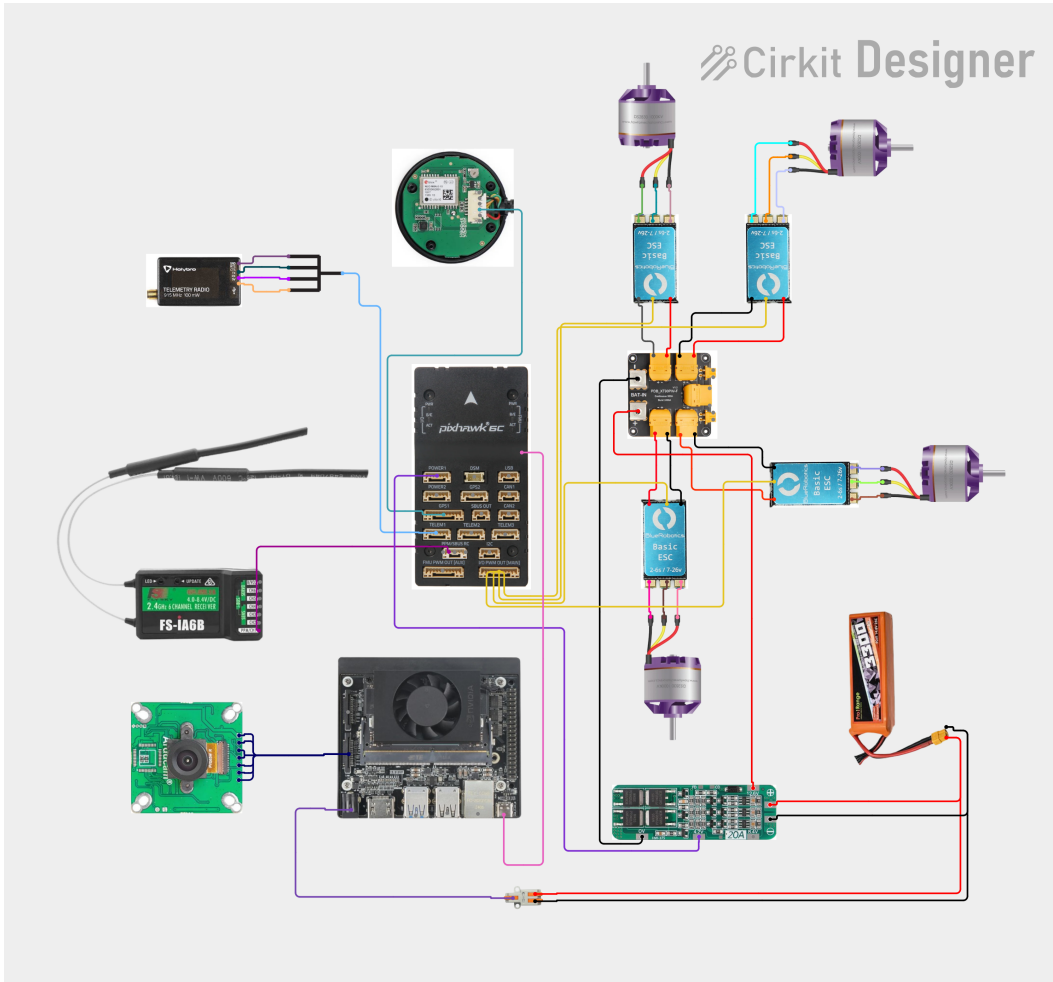


Figure 2: Exploded view showing modules, connector types, and wiring

4 Software Overview

4.1 Onboard Software

Pixhawk

PX4 Autopilot v1.14+ [1]: Open-source flight stack managing low-level flight control, sensor fusion, geofencing, and safety features. The Pixhawk comes pre-flashed and ready to use. To update the firmware to a newer version, see Section 5.1.

NVIDIA Jetson Orin Nano

ROS 2 Humble [2] with MAVROS [3]: communicates with the Pixhawk via `/dev/ttyTHS1`

Drone Brain: Repository with autonomous missions, tests and helper scripts. This repository is actively developed. For the current list of available missions, tests, and helpers, refer to the repository README and the `docs/` folder. See Section 5.3 for setup instructions.

Download: <https://github.com/autonomous-forest-drone/drone-brain>

4.2 Ground Control Software

QGroundControl (QGC)

QGC is the primary ground control station application. It is used for:

- Flashing and configuring Pixhawk firmware
- Sensor calibration and parameter management
- Mission planning and waypoint upload
- Real-time telemetry monitoring during flight
- Flight log download and analysis

Download: <https://qgroundcontrol.com>

Ground Control Station scripts

The `ground-control-station` repository runs on the monitoring computer. It is currently under active development. It includes an image receiver to display photos transmitted by the drone's `image-sender` helper. See Section 5.3 for setup instructions, and refer to the repository README for the current state of available scripts.

Download: <https://github.com/autonomous-forest-drone/ground-control-station>

5 Setup and Configuration

This section is written as an onboarding workflow. Complete the subsections in order. Figure 3 provides an overview of the full sequence.

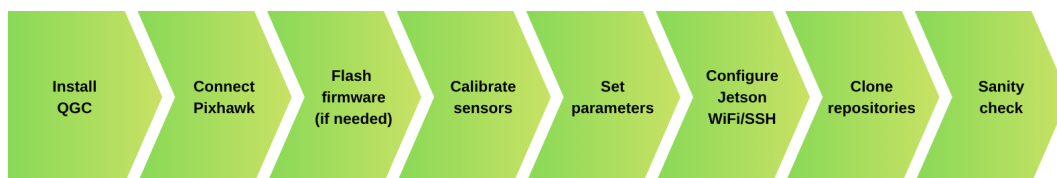


Figure 3: Setup workflow: onboarding sequence from QGC install to operational readiness

5.1 Setup Guide 1: QGroundControl and Pixhawk

Prerequisites:

- Pixhawk disconnected from propellers and main battery
- Micro-USB cable
- Ground control computer with internet access

Step 1 – Install QGroundControl

Download and install QGroundControl from <https://qgroundcontrol.com>.

Step 2 – Connect Pixhawk and verify detection

Connect Pixhawk to the computer by Micro-USB and open QGroundControl.

Step 3 – Flash or upgrade PX4 firmware (only if necessary)

The Pixhawk comes pre-flashed and ready to use. This step is only required if you are setting up a replacement unit, recovering from corrupted firmware, or upgrading to a newer PX4 stable release.

Go to **Vehicle Setup** → **Firmware**, select **PX4 Flight Stack – Stable Release**, then confirm flash.

When prompted, choose airframe **Holybro S500** (or **Generic Quadrotor X** if S500 is unavailable).

Step 4 – Run calibration sequence

In QGC, run the following in order:

1. **Accelerometer** – Vehicle Setup → Sensors → Accelerometer
2. **Compass** – Vehicle Setup → Sensors → Compass (perform outdoors, away from metal objects and electronics)
3. **Radio (RC)** – Vehicle Setup → Radio
4. **Flight Modes** – Vehicle Setup → Flight Modes
5. **ESC Calibration** – Vehicle Setup → Power → ESC Calibration (propellers removed)

Step 5 – Apply PX4 parameters

Open **Vehicle Setup** → **Parameters** and set all required values listed in Appendix C.

Step 6 – Save and verify configuration

Export parameters to a backup file and reboot Pixhawk once.

Important: Firmware flashing resets parameters to defaults. Always repeat Step 4 and Step 5 after reflashing. Detailed calibration actions are in Appendix E.

5.2 Setup Guide 2: Jetson Network and SSH Access

Prerequisites:

- Jetson powered and connected to Pixhawk via UART
- Same WiFi network available for Jetson and operator computer

Step 1 – Connect Jetson to internet

On the Jetson, run:

```

1 nmcli device wifi list
2 nmcli device wifi connect "NetworkName" password "yourpassword"
3 nmcli connection show

```

Step 2 – Find Jetson IP address

Run on the Jetson:

```

1 ip addr show

```

Record the IPv4 address for the active WiFi interface.

Step 3 – SSH from operator computer

```

1 ssh beetlesniffer@<jetson-ip-address>

```

Step 4 – Verify Pixhawk serial link

On the Jetson, run:

```

1 ls /dev/ttyTHS*

```

Optional (recommended): set a static IP for consistent SSH access.

```

1 nmcli connection modify "NetworkName" ipv4.addresses 192.168.1.100/24
2 nmcli connection modify "NetworkName" ipv4.method manual
3 nmcli connection up "NetworkName"

```

5.3 Setup Guide 3: Repository Clone, Update, and Version Control

Both repositories are actively developed. Use a pinned tag or commit for repeatable field deployments, and record the chosen versions in mission documentation.

Step 1 – Clone repositories (first-time setup)

On the Jetson:

```

1 git clone https://github.com/autonomous-forest-drone/drone-brain.git ~/drone-brain
2 cd ~/drone-brain
3 pip3 install -r requirements.txt

```

On the monitoring computer:

```

1 git clone https://github.com/autonomous-forest-drone/ground-control-station.git ~/ground-control-station
2 cd ~/ground-control-station
3 pip3 install -r requirements.txt

```

After cloning each repository, **read the README carefully and follow its setup instructions** before proceeding. The README is the authoritative source for any additional dependencies or configuration steps specific to that repository.

Step 2 – Update repositories (existing installations)

On each machine:

```

1 cd ~/drone-brain # or ~/ground-control-station
2 git fetch --all --tags
3 git pull origin main

```

Step 3 – Check out approved version

Use the exact version chosen by your team:

```
1 git checkout <tag-or-commit>
```

Step 4 – Run a quick sanity check

Run each repository's README quick-start commands and confirm no startup errors before flight-day operations.

6 Pre-Flight Checklist

Complete every item before every flight – no exceptions.

Physical Inspection

- ☐ All propellers installed correctly (CW/CCW per motor diagram, Appendix [D](#))
- ☐ No cracks or damage to frame, arms, or propellers
- ☐ All motor mount screws tight
- ☐ Battery secured with velcro straps, centred on frame
- ☐ Camera lens clean and facing forward
- ☐ GPS module upright and unobstructed
- ☐ No loose wires or connectors

Battery

- ☐ Fully charged: 16.4–16.8 V (4.1–4.2 V per cell)
- ☐ No swelling or physical damage
- ☐ XT60 connectors clean and undamaged

Environment

- ☐ Clear sky view (no dense overhead canopy blocking GPS)
- ☐ Wind below 3 m/s
- ☐ No rain or fog
- ☐ Temperature between 0–35°C
- ☐ 50 m safety perimeter clear of people and animals

Software and Telemetry (check in QGC)

- ☐ QGC connected and showing live telemetry
- ☐ GPS: 12+ satellites, 3D fix, HDOP below 1.5
- ☐ Battery voltage 15.5 V or above
- ☐ All sensors showing green (no red indicators)
- ☐ RC transmitter on and all channels responding in QGC
- ☐ Geofence configured (100 m horizontal, 50 m vertical)
- ☐ Flight mode set correctly for the planned mission

7 Operation

7.1 Startup Procedure

1. Place the drone in an open outdoor area
2. Power on the RC transmitter
3. Connect the main battery to the XT60 Y-splitter
4. Wait for the Pixhawk to boot (LED flashes during boot, solid when ready)
5. Wait approximately 60 seconds for the Jetson to boot
6. Connect QGC via the telemetry radio link
7. **Wait at least 5 minutes** before arming – the barometer and GPS need time to stabilise

7.2 RC Transmitter Guide

The FlySky FS-i6 transmitter provides manual control and safety override. It always takes priority over autonomous commands.

Stick layout (Mode 2 – default):

- **Left stick vertical:** Throttle (up = climb)
- **Left stick horizontal:** Yaw (rotate left/right)
- **Right stick vertical:** Pitch (forward/backward)
- **Right stick horizontal:** Roll (tilt left/right)

Switches:

- **Channel 5:** Flight mode selector (3-position switch)
- **Channel 6:** Arming switch

7.3 Arming and Disarming

Arming:

1. Set throttle stick to full down
2. Move rudder stick to full right
3. Hold for 2 seconds
4. Motors begin spinning at idle – QGC shows **ARMED**

Disarming:

1. Set throttle stick to full down
2. Move rudder stick to full left
3. Hold for 2 seconds
4. Motors stop – QGC shows **DISARMED**

7.4 Flight Modes

Mode	Description	Typical Use
Manual	No stabilisation	Not recommended
Stabilise	Auto-levels when sticks are centred	Manual flight practice
Altitude Hold	Holds altitude automatically	Basic altitude testing
Position	GPS position hold + altitude hold	GPS testing, stationary hover
Offboard	Jetson controls via MAVROS	Autonomous scripted mission
RTL	Returns to launch automatically	Emergency or end of mission

Modes are selected via the Channel 5 switch. Configure the switch positions in **QGC** → **Vehicle Setup** → **Flight Modes**.

Note: Switching to Stabilise or Altitude Hold at any time immediately cancels Offboard mode and returns control to the RC transmitter.

7.5 Using QGroundControl

7.5.1 Connecting to the Drone

- **Via telemetry radio:** QGC connects automatically when the ground radio is plugged in and the drone is powered
- **Via USB:** Connect the Pixhawk with a Micro-USB cable; QGC detects it automatically

7.5.2 Main Flight Screen

- **Top bar:** GPS count, battery voltage, flight mode, armed status, telemetry signal strength

- **Map view:** Drone position, home point, geofence, mission waypoints
- **Attitude indicator:** Live pitch and roll
- **Altitude and speed:** Current values in real time

7.5.3 Planning a Mission (Waypoints)

1. Click **Plan** (map icon) in the left toolbar
2. Click on the map to place waypoints
3. Set altitude for each waypoint (5–10 m AGL recommended)
4. Click **Upload** to send the mission to the Pixhawk
5. Return to the **Fly** view to monitor execution

7.5.4 Viewing and Changing Parameters

1. Vehicle Setup → Parameters
2. Use the search bar to find a parameter by name
3. Click the parameter, enter the new value, click **Save**
4. Reboot the Pixhawk if QGC prompts for a reboot

All critical parameters are listed in Appendix C.

7.5.5 Configuring the Geofence

1. Plan view → Fence (fence icon in toolbar)
2. Draw an inclusion zone around the flight area
3. Set maximum altitude (50 m recommended)
4. Click **Upload**

7.5.6 Downloading Flight Logs

1. Vehicle Setup → Analyse Tools → Log Download
2. Select the log file by date and time
3. Click **Download**
4. Upload the `.ulg` file to <https://logs.px4.io> for online analysis

7.6 Running a Mission or Test

All scripts in `drone-brain` follow the same two-terminal pattern. All terminals must be SSH sessions on the Jetson.

Terminal 1 – start MAVROS (required for every mission or test):

```
1 ros2 launch mavros px4.launch fcu_url:=/dev/ttyTHS1:115200
```

Terminal 2 – run a mission, test, or helper:

```
1 # Run a mission
2 python3 ~/drone-brain/missions/<mission>.py
3
4 # Run a pre-flight test
5 python3 ~/drone-brain/test/<test>.py
6
7 # Run a helper (alongside a mission, in a separate terminal)
8 python3 ~/drone-brain/helpers/<helper>.py
```

Currently available scripts (check the repository for additions):

Type	Name	Description
Test	hover-test	Basic hover to verify arming and altitude hold
Helper	telemetry-monitoring	Prints live telemetry to the terminal
Helper	image-sender	Captures a frame from the Arducam IMX219 via GStreamer, resizes to 128×128, and transmits it to the ground station in 128-byte chunks over MAVROS Tunnel messages every 20 seconds

For the full and up-to-date list of available missions, tests, and helpers, read the repository README and the docs/ folder: <https://github.com/autonomous-forest-drone/drone-brain>

Emergency override: Switch to Stabilise or Altitude Hold on the RC transmitter at any time to cancel Offboard mode and take manual control.

Recommended: use `tmux` to keep sessions alive if SSH disconnects:

```
1 # Start a named session
2 tmux new -s drone
3
4 # Split into two panes: Ctrl+B then % (or ")
5 # Detach (session keeps running): Ctrl+B then D
6 # Reconnect later:
7 tmux attach -t drone
```

7.7 Running Ground Control Station Scripts

On the monitoring computer, start the relevant script from the `ground-control-station` repository before or during a mission.

The repository is under active development. Refer to its README for the current list of scripts and usage instructions: <https://github.com/autonomous-forest-drone/ground-control-station>

The `image-receiver` script will receive and display the images transmitted by the drone's `image-sender` helper.

7.8 In-Flight Monitoring

Keep QGC open and watch these values throughout every flight:

Parameter	Normal Range	Action if Out of Range
Battery voltage	14.4–16.8 V	Land immediately below 14.0 V
Battery remaining	30–100%	RTL auto-triggers at 20%
GPS satellites	12 or more	Land if count drops below 8
HDOP	below 1.5	Land if above 2.5
Speed (autonomous)	below 3 m/s	Check Offboard mode parameters
Altitude deviation	within ± 2 m of plan	Investigate if larger

Land immediately if any of the following occur:

- Battery below 20%
- “GPS lost” warning in QGC
- Unusual vibration, noise, or erratic movement
- Telemetry lost for more than 10 seconds

7.9 Landing Procedures

Manual landing:

1. Switch RC to Stabilise or Altitude Hold mode
2. Position drone over the landing zone
3. Reduce throttle gradually – target descent rate 0.3–0.5 m/s
4. Once on the ground, lower throttle to minimum
5. Disarm (throttle down + rudder left, hold 2 seconds)

Autonomous landing: The mission script lands automatically at the end of the waypoint sequence. To trigger an early return, switch the flight mode to RTL via the Channel 5 switch or send the command from QGC. The drone will fly back to the launch point, descend, land, and auto-disarm 2 seconds after touchdown.

7.10 Post-Flight Shutdown

1. Confirm motors are disarmed and fully stopped
2. Wait 10 seconds for components to cool
3. Disconnect the main battery
4. Power off the RC transmitter
5. Download flight logs from QGC if needed
6. Inspect propellers and frame for any damage
7. Store the LiPo battery in a fireproof LiPo bag

8 Troubleshooting

8.1 Common Issues

Symptom	Likely Cause	Solution
Drone will not arm	Pre-flight check failure	Check QGC for red indicators; recalibrate sensors
No GPS fix	Poor satellite reception	Move to open area; wait up to 10 minutes
ESCs beeping continuously	No signal or not armed	Check wiring; recalibrate ESCs
Altitude drifts	Barometer not warmed up	Wait 5 minutes after power-on
Unstable hover or oscillations	Vibration or PID tuning	Check vibration below 30 m/s ² ; reset PIDs to default
One side lifts first	Motor mapping incorrect	Verify motor assignments in QGC Actuators tab
Drone will not take off	Low battery or weak motors	Charge battery; confirm hover throttle around 60%
Compass error	Magnetic interference	Recalibrate outdoors; move GPS mast away from power cables
Telemetry lost	Radio or range issue	Check antenna orientation; reduce distance
MAVROS fails to connect	UART device not found or no permission	Check <code>/dev/ttyTHS1</code> exists; add user to dialout group (<code>sudo usermod -aG dialout \$USER</code> , then log out and back in)
Mission script errors	Missing dependency or wrong path	Check <code>requirements.txt</code> installed; verify script path

8.2 Component-Specific Diagnostics

8.2.1 Pixhawk

In QGC, go to **Analyse Tools** → **MAVLink Inspector** to monitor live MAVLink messages:

- **VIBRATION** – vibration_x/y/z should be below 30 m/s²
- **GPS_RAW_INT** – satellite count, fix type, HDOP
- **BATTERY_STATUS** – voltage, current, remaining percentage
- **ATTITUDE** – roll, pitch, yaw
- **SERVO_OUTPUT_RAW** – motor PWM values (should be balanced within $\pm 20 \mu\text{s}$)

8.2.2 GPS Module (Holybro M10)

If GPS does not achieve a 3D fix:

- Move to a clear outdoor area with unobstructed sky

- Wait up to 10 minutes for cold-start acquisition
- Verify the GPS antenna is upright and unobstructed
- Check the GPS cable connection at the Pixhawk GPS port
- Confirm HDOP below 1.5 in QGC before arming

8.2.3 RC Receiver (FlySky FS-iA6B)

If RC channels are not responding in QGC:

- Confirm the transmitter is powered on
- Check the receiver LED: solid = bound; flashing = unbound
- To rebind: hold the bind button on the receiver while powering on the transmitter in bind mode (see the FS-i6 user manual [6] for the bind mode procedure)
- Confirm receiver is in PPM mode: 2 LED flashes on power-on = PPM; 1 flash = PWM
- Recalibrate RC in QGC: Vehicle Setup → Radio

8.2.4 Telemetry Radio (RFD900x / SiK)

If QGC shows no telemetry link:

- Check antenna connections on both the ground and air units
- Verify the ground unit COM port is selected in QGC
- Test at short range first (<10 m)
- Both units must be on the same frequency band (both 915 MHz or both 433 MHz)
- LED status: green = linked; red = searching

8.2.5 Jetson and MAVROS

```

1 # Check MAVROS connection and flight controller state
2 ros2 topic echo /mavros/state
3
4 # Check camera is publishing frames
5 ros2 topic hz /camera/image_raw
6
7 # Jetson system and GPU utilisation
8 tegrastats
9
10 # Check UART device is present
11 ls /dev/ttyTHS1
12
13 # Fix UART permissions if access is denied
14 sudo usermod -aG dialout $USER

```

8.2.6 Battery and Power System

- **Check cell voltages:** Use a LiPo voltage checker on the balance connector. Fully charged = 4.2 V per cell (16.8 V total); minimum = 3.5 V per cell (14.0 V total)
- **If the drone shuts down mid-flight:** Check minimum cell voltage; inspect XT60 connectors for looseness or corrosion

8.3 Emergency Procedures

Loss of control:

1. Switch to RTL mode using the Channel 5 switch or trigger it from QGC
2. If RTL does not engage, switch to Stabilise mode and land manually
3. If RC is unresponsive, the drone enters automatic failsafe RTL after 10 seconds

GPS lost during flight:

1. Drone switches automatically to Altitude Hold (maintains height but may drift)
2. Immediately switch to Altitude Hold or Stabilise mode on the RC
3. Land at the nearest safe location
4. Do not re-enter Position or Offboard mode without a GPS fix

Low battery warning:

1. QGC shows red battery alert and audio warning
2. Drone triggers automatic RTL at 20%
3. Do not override – allow it to return and land
4. If below 10%, land at the nearest safe spot immediately

Crash or collision:

1. Drone auto-disarms on heavy impact
2. Approach cautiously (propellers may still be sharp even when stopped)
3. Disconnect the battery immediately
4. Inspect all components before attempting another flight
5. Download the crash log from QGC for analysis

9 Safety Guidelines

9.1 Hazards

Mechanical:

- Rotating propellers can cause severe lacerations – maintain safe distance during operation
- Carbon fibre frame edges are sharp – handle with care
- A 2.3 kg drone falling from height can cause serious injury

Electrical:

- LiPo batteries can catch fire if overcharged, punctured, or short-circuited
- ESCs draw up to 40 A – never short power cables
- Reversed battery polarity causes instant and irreversible component damage

Environmental:

- Risk of collision with trees, buildings, people, or power lines
- Flyaway risk in high wind or near GPS-blocking structures
- Camera recording may raise privacy concerns – handle data per GDPR

9.2 Safe Operating Rules

- Establish a 50 m safety perimeter before arming
- Keep the RC transmitter in hand throughout every flight
- Maintain visual line of sight at all times
- Never fly over people, roads, vehicles, or buildings
- Have a fireproof LiPo bag or extinguisher available when charging
- Disconnect the battery immediately after landing
- Store LiPo batteries in a fireproof LiPo bag when not in use
- Allow motors to cool for at least 5 minutes before handling after a flight

10 Legal and Regulatory Compliance

10.1 EU Drone Regulations (as of 2026)

[4]

- **Registration:** Drones over 250 g must be registered with the national aviation authority
- **Operator certificate:** A2 certificate required for autonomous operations near people
- **Maximum altitude:** 120 m AGL
- **No-fly zones:** Within 5 km of airports; military installations; nuclear facilities; nature reserves (permit required)
- **Privacy:** Camera data must comply with GDPR
- **Insurance:** Liability insurance recommended for drones over 900 g

10.2 Sweden-Specific Requirements

- Register with Transportstyrelsen (Swedish Transport Agency) [5]
- Follow Luftfartsverket (Swedish CAA) regulations
- Forest operations may require landowner permission
- Check local restrictions before each operation

10.3 Disclaimer

This system is an academic research prototype and carries no commercial warranty or certification. Operators are solely responsible for safe operation, regulatory compliance, and any injury or damage resulting from use.

© 2026 Elina Rosato, Danielis Maizelis, Kristianstad University. Custom code is released under the Apache 2 License. See the project repository for details.

A Glossary

Term	Definition
AGL	Above Ground Level
CCW	Counter-Clockwise
CW	Clockwise
EKF	Extended Kalman Filter (sensor fusion algorithm)
ESC	Electronic Speed Controller
GCS	Ground Control Station
GNSS	Global Navigation Satellite System
HDOP	Horizontal Dilution of Precision (GPS accuracy metric; below 1.5 is good)
IMU	Inertial Measurement Unit
LiPo	Lithium Polymer battery
MAVLink	Micro Air Vehicle communication protocol
PDB	Power Distribution Board
PID	Proportional-Integral-Derivative (flight control algorithm)
PPM	Pulse Position Modulation (RC signal format)
PWM	Pulse Width Modulation (motor control signal)
ROS	Robot Operating System
RTL	Return to Launch
UART	Universal Asynchronous Receiver-Transmitter

B Technical Specifications

Parameter	Value
Flight Performance	
Maximum flight time	12–15 minutes
Maximum speed (autonomous)	3 m/s
Maximum speed (manual)	5 m/s
Maximum operating altitude	50 m AGL
Wind resistance	below 3 m/s sustained
Physical	
Total weight	2.3 kg (including battery)
Frame diagonal	500 mm
Overall dimensions	550 × 550 × 200 mm
Propeller size	10 × 4.5 inches
Power	
Battery	4S LiPo 14.8 V 5000 mAh
Voltage range	12.8 V (empty) to 16.8 V (full)
Peak current draw	60 A (all motors)
Sensors	
GPS	Holybro M10 (GPS + GLONASS)
Barometer	MS5611 (± 10 cm accuracy)
Camera	Ardurcam IMX219, 8 MP, 62° FOV, CSI, fixed focus
Communication	
Telemetry	915 MHz or 433 MHz, up to 1 km
Jetson–Pixhawk	UART serial, <code>/dev/ttyTHS1:115200</code> , MAVLink 2.0
RC	2.4 GHz AFHDS
Computing	
Companion computer	NVIDIA Jetson Orin Nano 8 GB
GPU	1024-core NVIDIA Ampere
RAM	8 GB
Storage	64 GB eMMC + microSD

C PX4 Parameter Reference

Set these parameters after a firmware flash via **QGC** → **Vehicle Setup** → **Parameters**.

Parameter	Value	Description
SYS_AUTOSTART	4014	Holybro S500 airframe
EKF2_HGT_REF	0	Use barometer for altitude reference
EKF2_BARO_NOISE	6.0	Barometer noise level
MPC_THR_HOVER	0.60	Hover throttle (60%)
COM_ARM_GPS_HDOP	1.5	Maximum HDOP for arming
EKF2_GPS_CTRL	7	GPS fusion control
COM_ARM_EKF_POS	0.5	EKF position check threshold
COM_DISARM_LAND	2.0	Auto-disarm 2s after landing
COM_RC_LOSS_T	1.0	RC loss failsafe timeout (seconds)
GF_MAX_HOR_DIST	100	Geofence horizontal limit (m)
GF_MAX_VER_DIST	50	Geofence vertical limit (m)
BAT_CRIT_THR	0.20	Critical battery threshold (20%)
BAT_LOW_THR	0.30	Low battery warning (30%)
SER_TEL2_BAUD	115200	TELEM2 baud rate for Jetson UART link

D Motor Configuration and Propeller Diagram

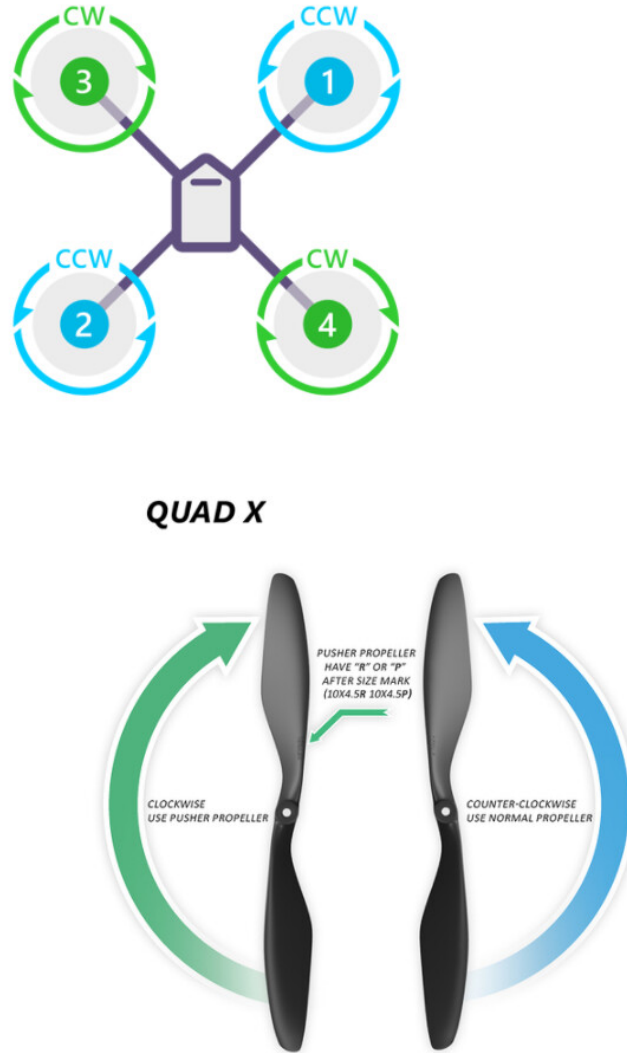


Figure 4: Motor positions, rotation directions, and MAIN OUT port assignments

Motor	Position	Direction	MAIN OUT
M1	Front-Right	CCW	1
M2	Back-Left	CW	2
M3	Front-Left	CW	3
M4	Back-Right	CCW	4

Propeller identification: CCW propellers are marked with “L” on the hub. CW propellers are unmarked or marked “R”. Mixing up CW and CCW propellers will cause the drone to flip on takeoff.

E Calibration Procedures

Accelerometer (6-Position)

1. QGC → Vehicle Setup → Sensors → Accelerometer
2. Level surface, nose pointing forward → click **Level Horizon**
3. Nose down, tail up 90° → click **Nose Down**
4. Nose up → click **Nose Up**
5. Right side down → click **Right Side Down**
6. Left side down → click **Left Side Down**
7. Upside down → click **Back**

Compass

1. Perform outdoors, at least 2 m from metal objects and electronics
2. QGC → Vehicle Setup → Sensors → Compass
3. Follow the on-screen instructions: rotate the drone through pitch, roll, and yaw axes
4. Complete all coloured segments in the sphere visualisation

ESC Calibration

1. Remove all propellers
2. Disconnect main battery
3. QGC → Vehicle Setup → Power → ESC Calibration → Start
4. Move throttle stick to maximum on the RC transmitter
5. Connect the main battery – ESCs emit confirmation beeps
6. Move throttle stick to minimum – ESCs emit a second confirmation
7. Disconnect battery; calibration is complete

RC Radio Calibration

1. QGC → Vehicle Setup → Radio
2. Move all sticks to their full extents (all four corners)
3. Move all switches through all positions
4. Click **Next** to complete

F Maintenance Schedule

Before Every Flight

- Visual inspection for cracks, loose screws, or damaged propellers
- Check propeller tightness
- Check battery voltage and condition

After Every 10 Flights

- Re-tighten all screws (motors, frame, propellers)
- Spin-test motors by hand – smooth with no grinding indicates healthy bearings
- Clean camera lens
- Inspect carbon fibre arms for delamination or micro-cracks

After Every 50 Flights

- Replace propellers (they degrade with use)
- Inspect ESC solder joints for cold joints or cracking
- Recalibrate compass and accelerometer
- Update Pixhawk firmware if a new stable release is available

Annually

- Replace LiPo battery (capacity degrades with charge cycles)
- Replace motor bearings if grinding is detected
- Full system inspection and test flight before field operations

References

- [1] PX4 Development Team, *PX4 Autopilot User Guide*, <https://docs.px4.io>, 2024.
- [2] Open Robotics, *ROS 2 Humble Hawksbill Documentation*, <https://docs.ros.org/en/humble/>, 2024.
- [3] MAVROS Contributors, *MAVROS – MAVLink extendable communication node for ROS 2*, <https://github.com/mavlink/mavros>, 2024.
- [4] European Union Aviation Safety Agency, *Easy Access Rules for Unmanned Aircraft Systems*, EASA, 2024.
- [5] Transportstyrelsen, *Regulations for UAS Operations in Sweden*, <https://transportstyrelsen.se/dronare>, 2024.
- [6] FlySky, *FS-i6 2.4G 6CH AFHDS RC Transmitter User Manual*, FlySky Model Co., Ltd.
- [7] Holybro, *S500 V2 Development Kit User Guide*, <https://holybro.com/products/s500-v2-development-kit>, 2024.